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ORF307: Optimization

Midterm-1 Preparation Exercises

Fall Semester 2025

At the end of each lecture slide, we have listed relevant chapters from the reference books. *All* exercises within those chapters are excellent practice for Midterm 1. To guide your preparation efforts, we also provide a list of some *representative* exercises from those chapters:

1 Least squares

From the [VMLS] book check the following exercises:

Chapter 12: 12.1, 12.4, 12.10, 12.15

Chapter 13: 13.3, 13.5, 13.7, 13.9

Chapter 15: 15.1, 15.4

Chapter 16: 16.3, 16.4, 16.9, 16.7, 16.10, 16.12

Chapter 17: 17.1, 17.4, 17.6

2 Linear Optimization

From the [LO] book check the following exercises:

Chapter 1: 1.3, 1.4, 1.6, 1.8, 1.10, 1.12, 1.14, 1.15

References

[VMLS]: S. Boyd, L. Vandenberghe: Introduction to Applied Linear Algebra – Vectors, Matrices, and Least Squares;

Available: <https://web.stanford.edu/~boyd/vmls/> .

[LO]: D. Bertsimas & J. Tsitsiklis: Introduction to Linear Optimization.

Available: Princeton Controlled Digital Lending .